LEMON Open lab

2026.11.06

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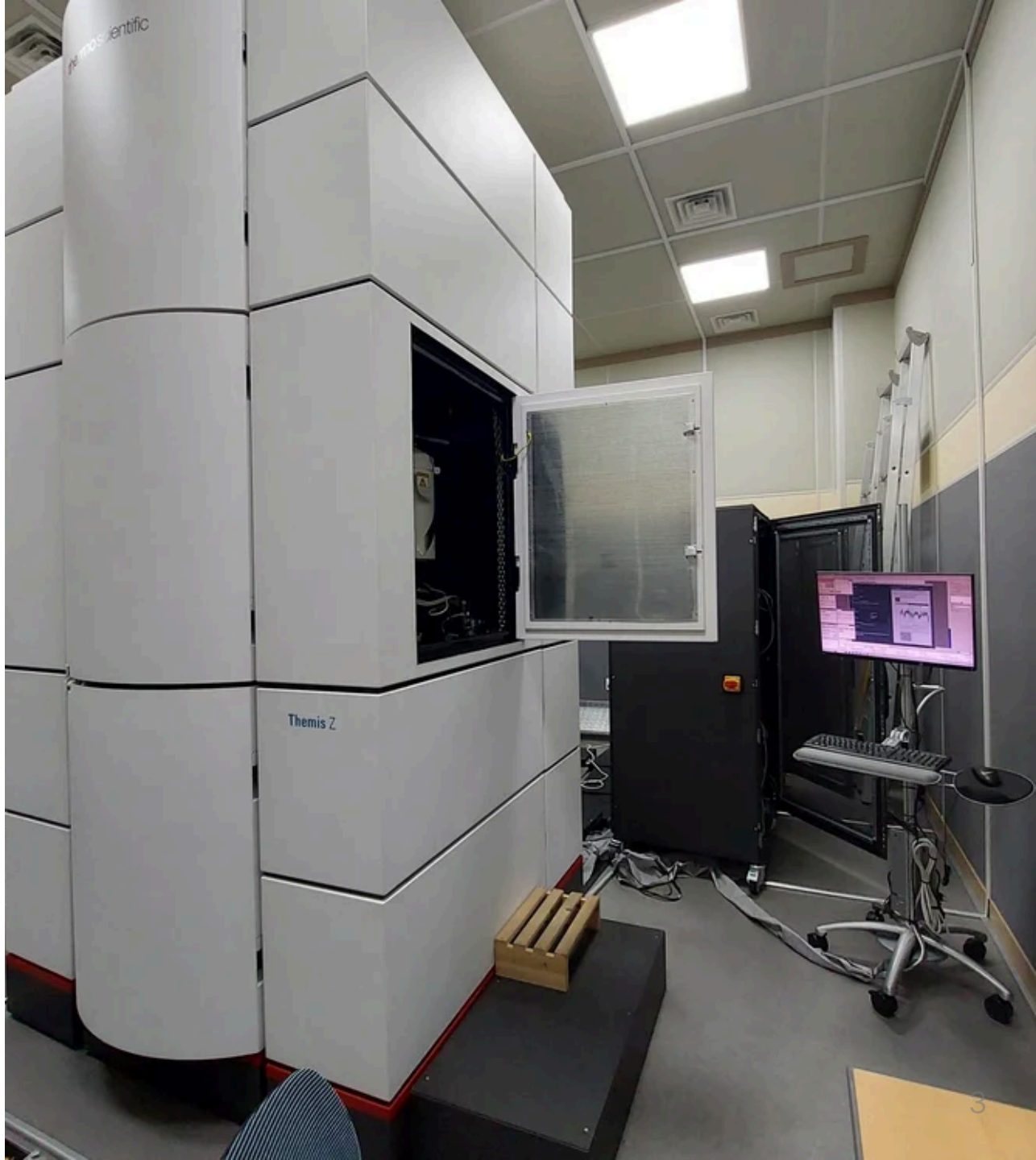
(Scanning) Transmission Electron Microscopes

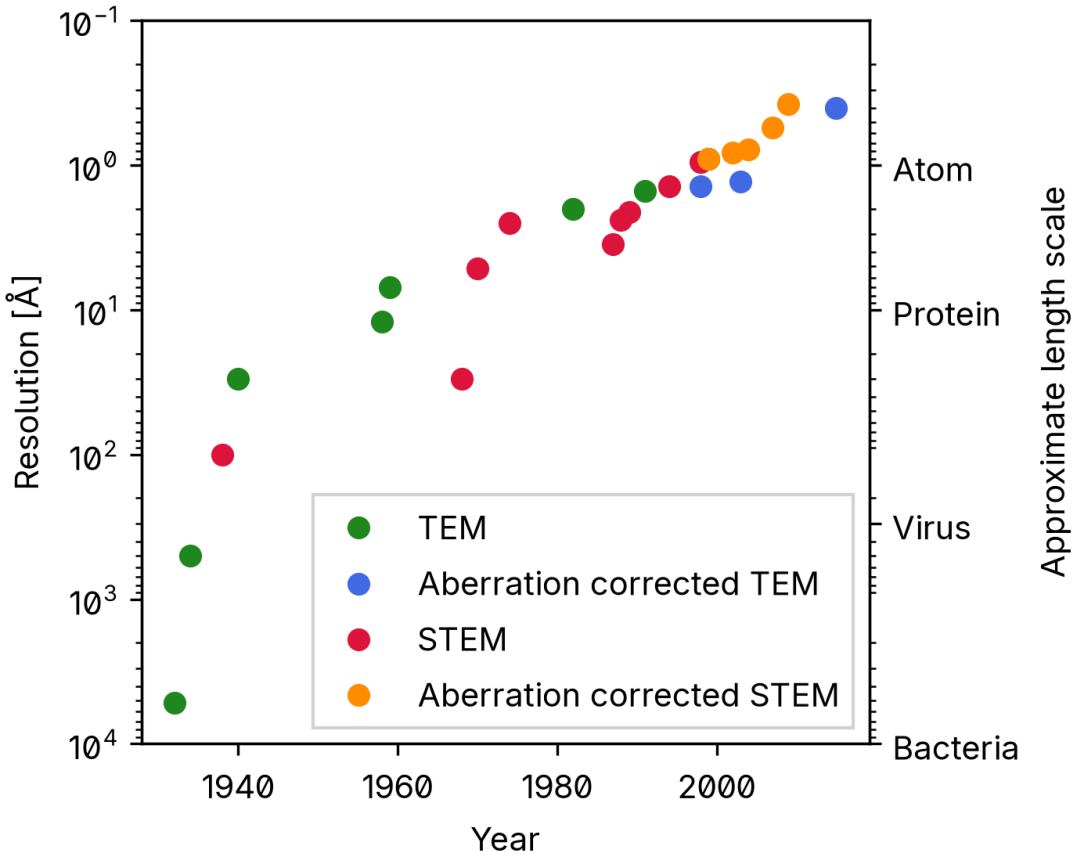
서울대학교 신소재공동연구소 Themis-Z, ...



(Scanning) Transmission Electron Microscopes

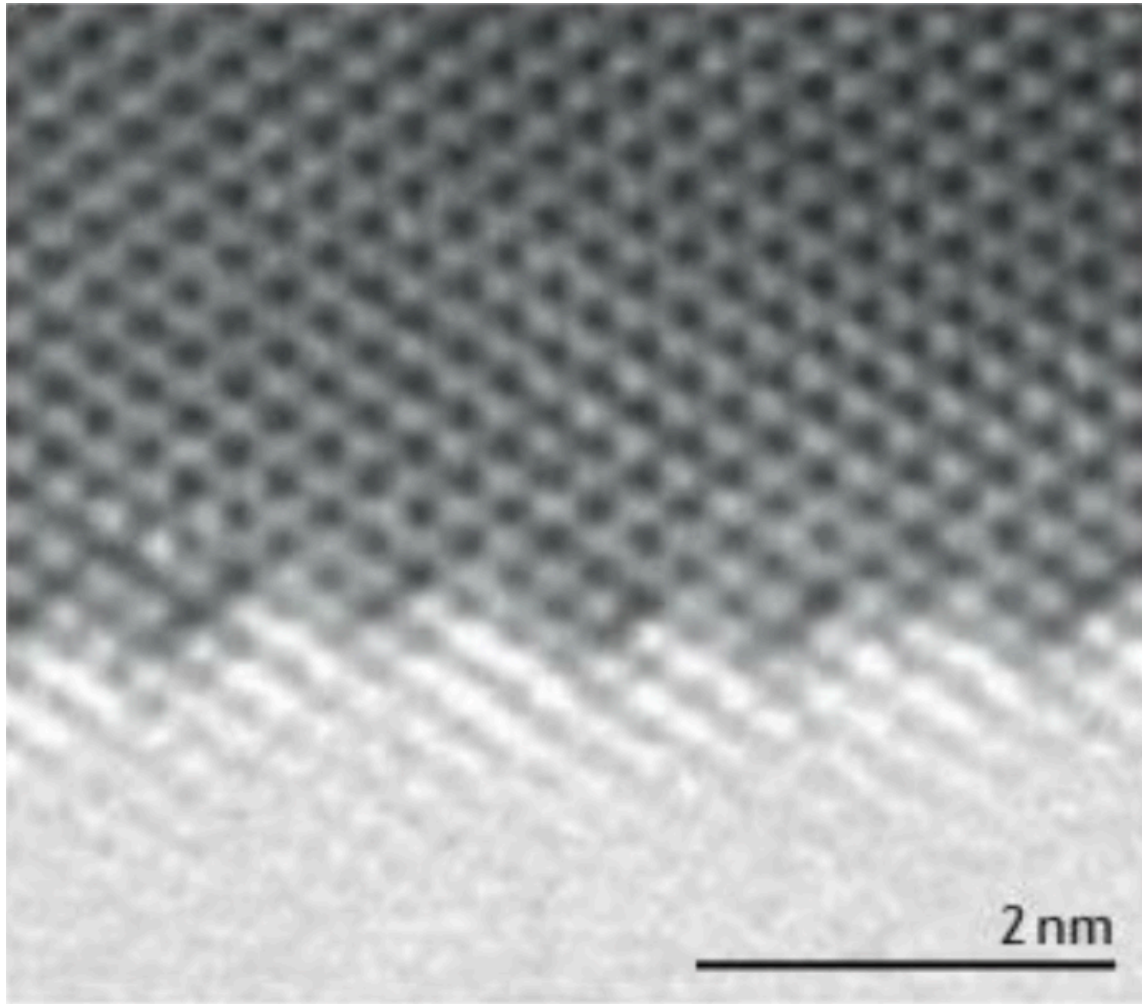
Cornell Spectra 300, ...

Transmission Electron Microscopy

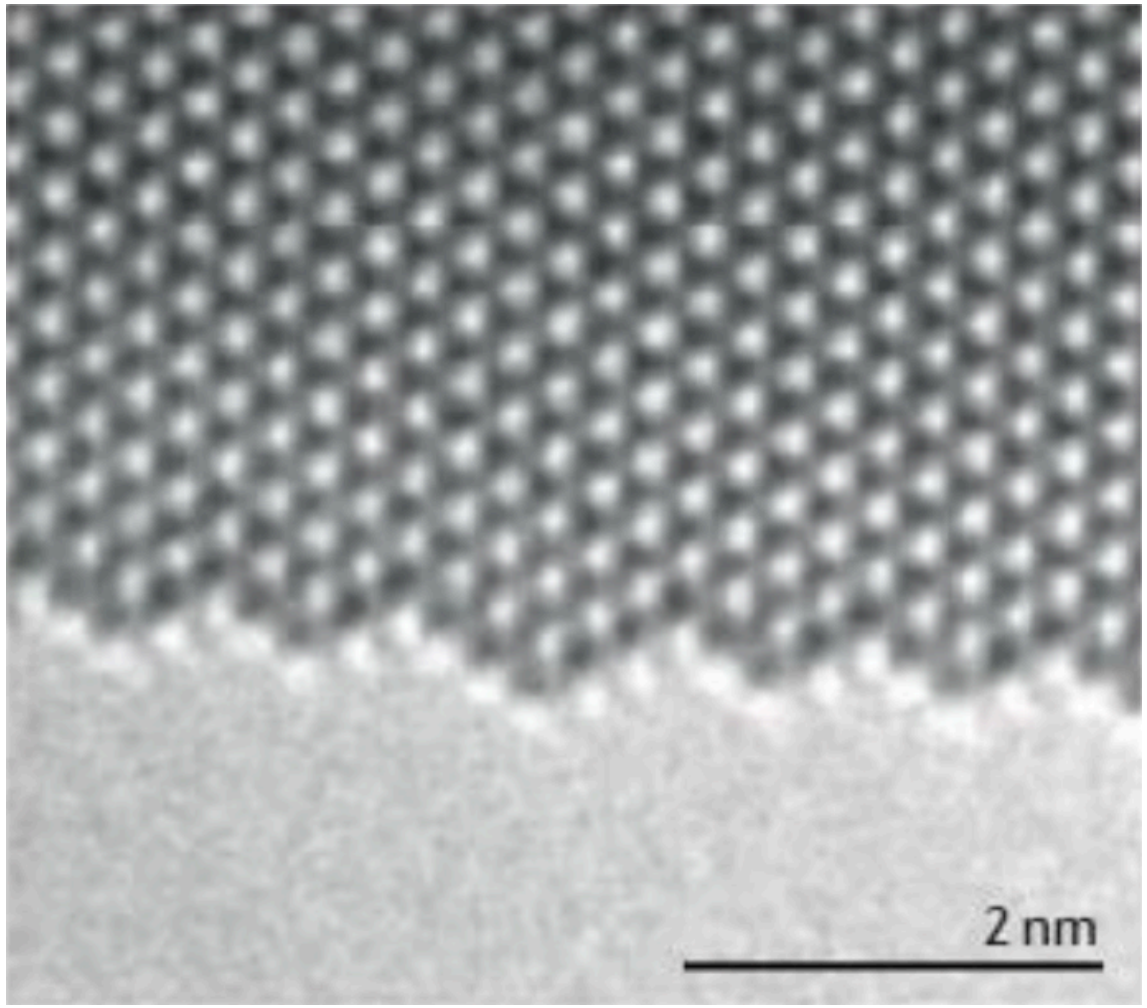


Atomic resolution microscopy: 1~0.1 Å

Conventional TEM (CTEM)

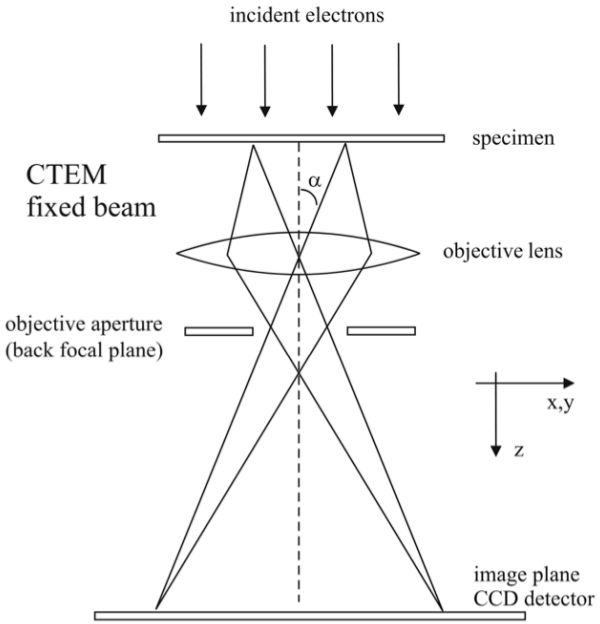


Cs-Corrected CTEM

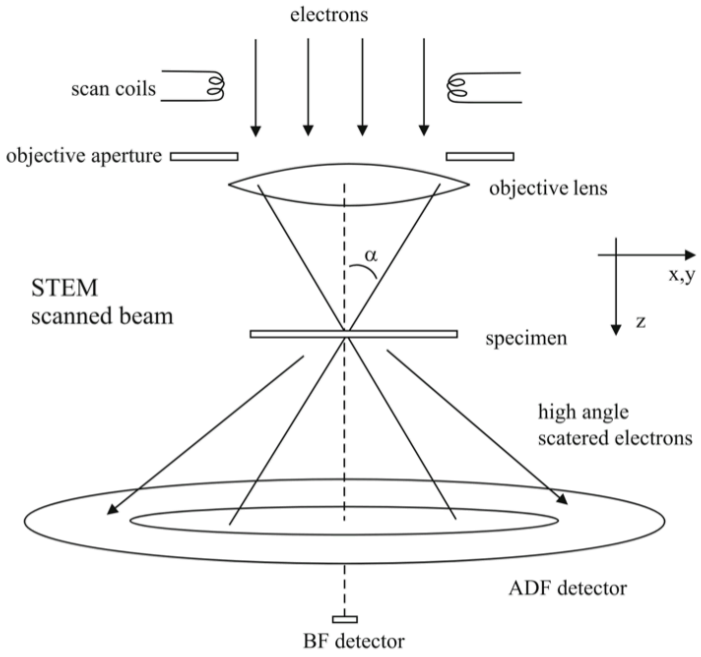


Scanning TEM (STEM)

- 수차, 시편 두께, 초점 거리 등에 덜 민감 → 고해상도 이미지
- Z-contrast, strain, chemical analysis 등 새로운 분석

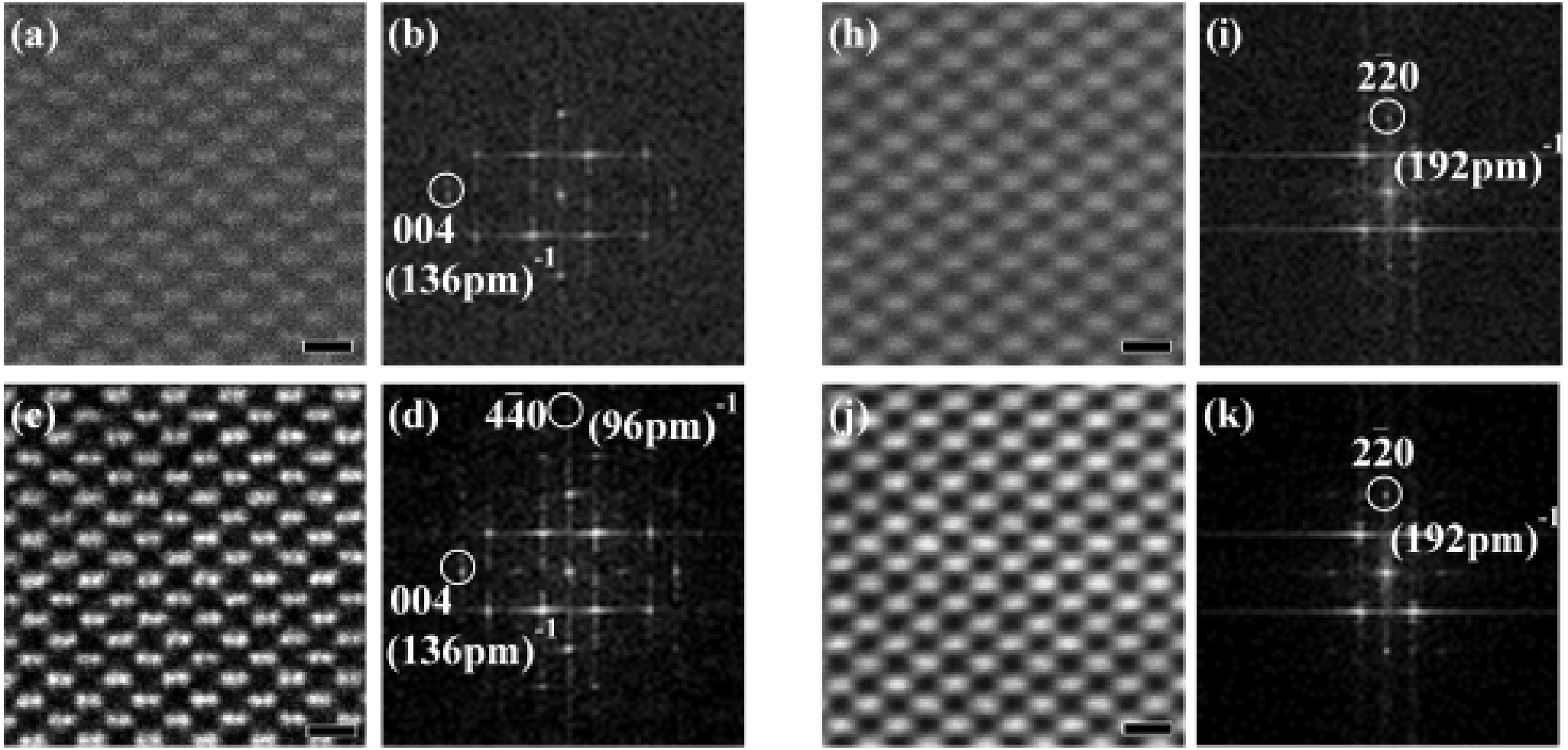


← CTEM

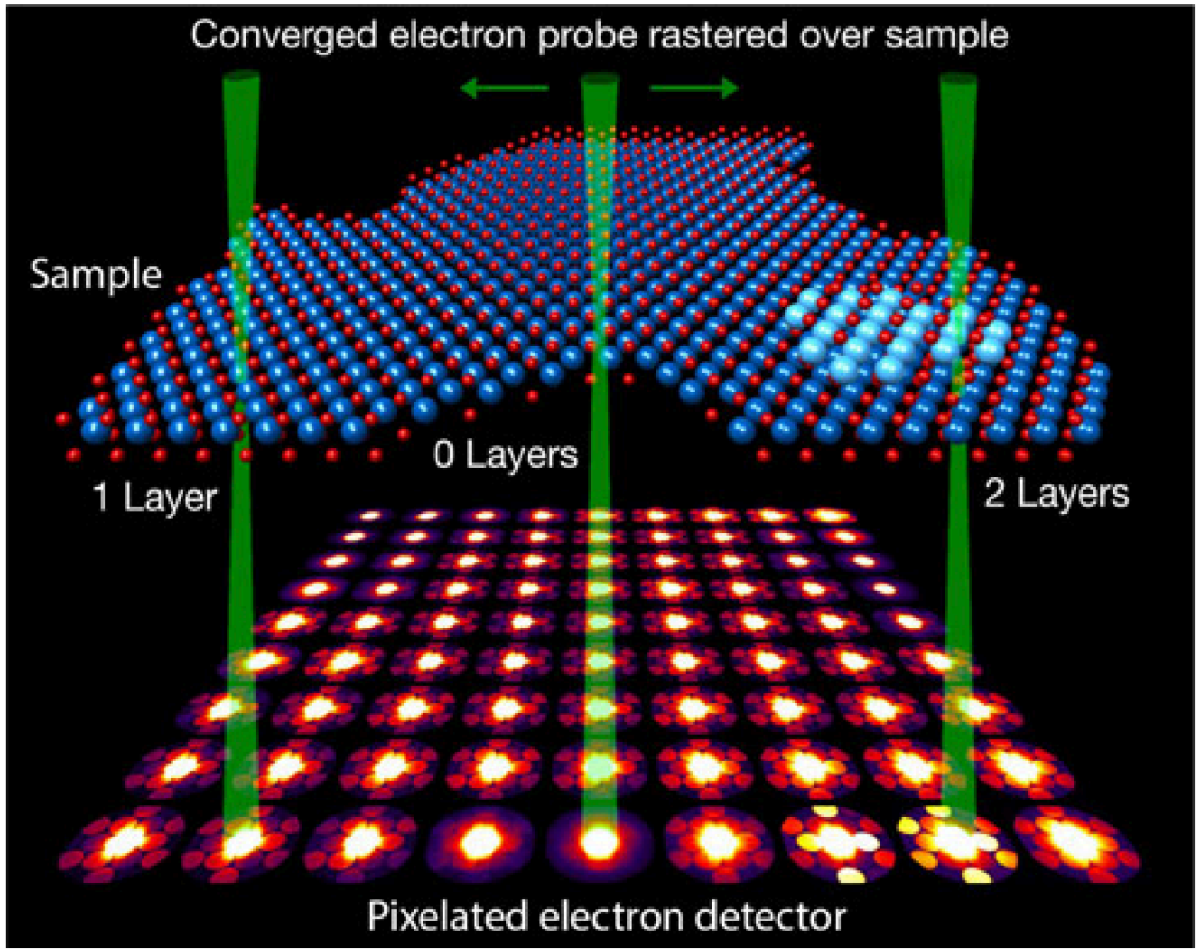


← STEM

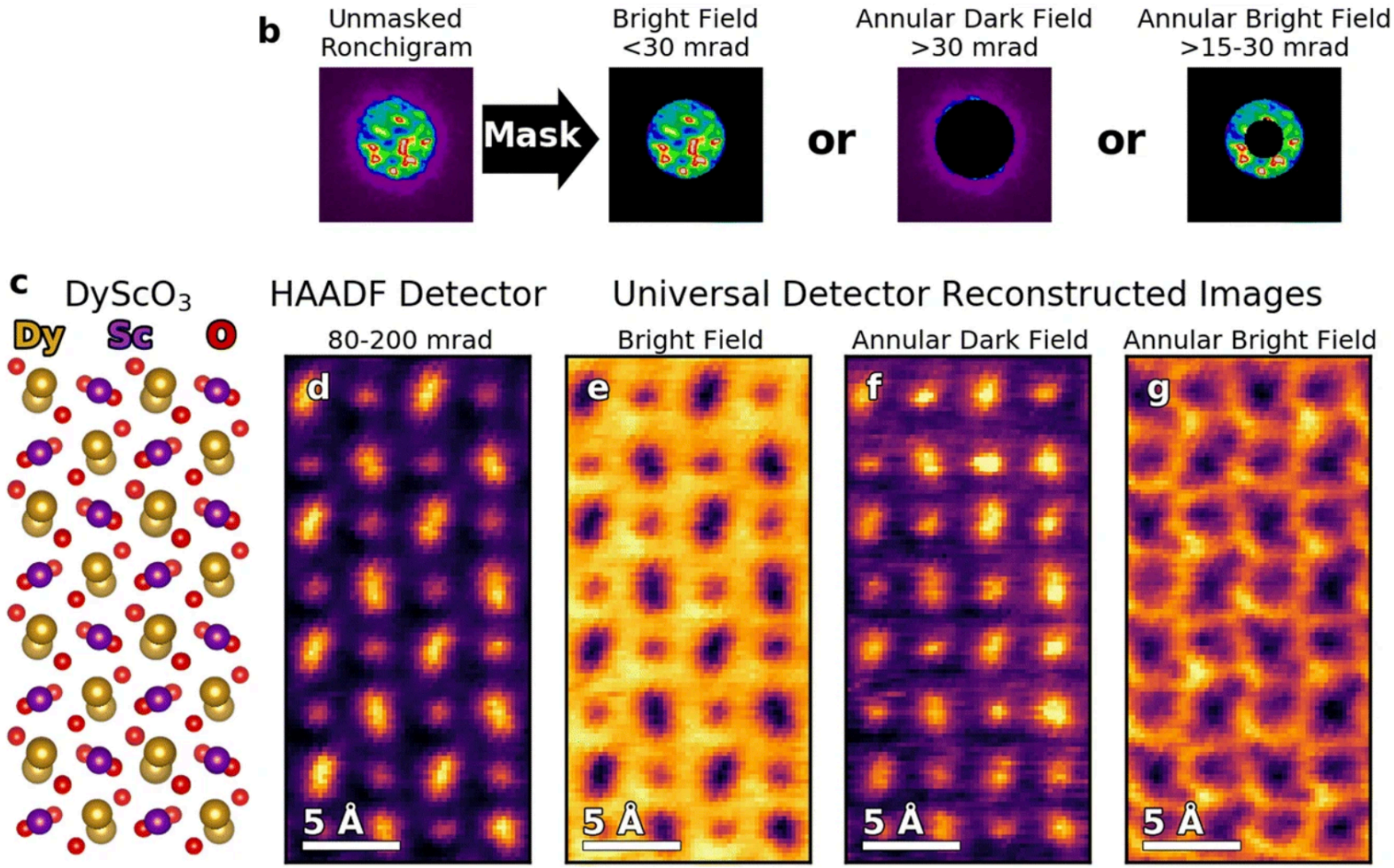
Cs-Corrected STEM



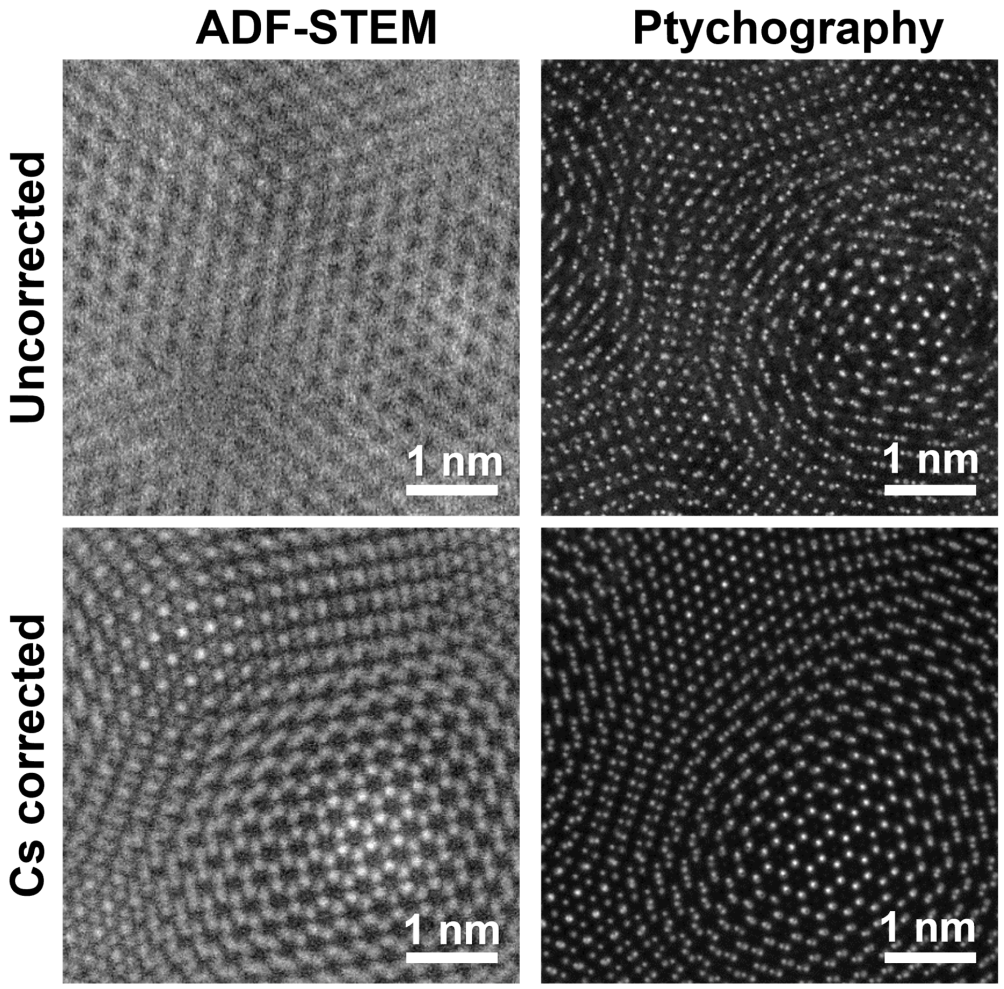
4D-STEM


Converged electron probe rastered over sample
Sample
0 Layers
1 Layer
2 Layers
Pixelated electron detector

4D-STEM

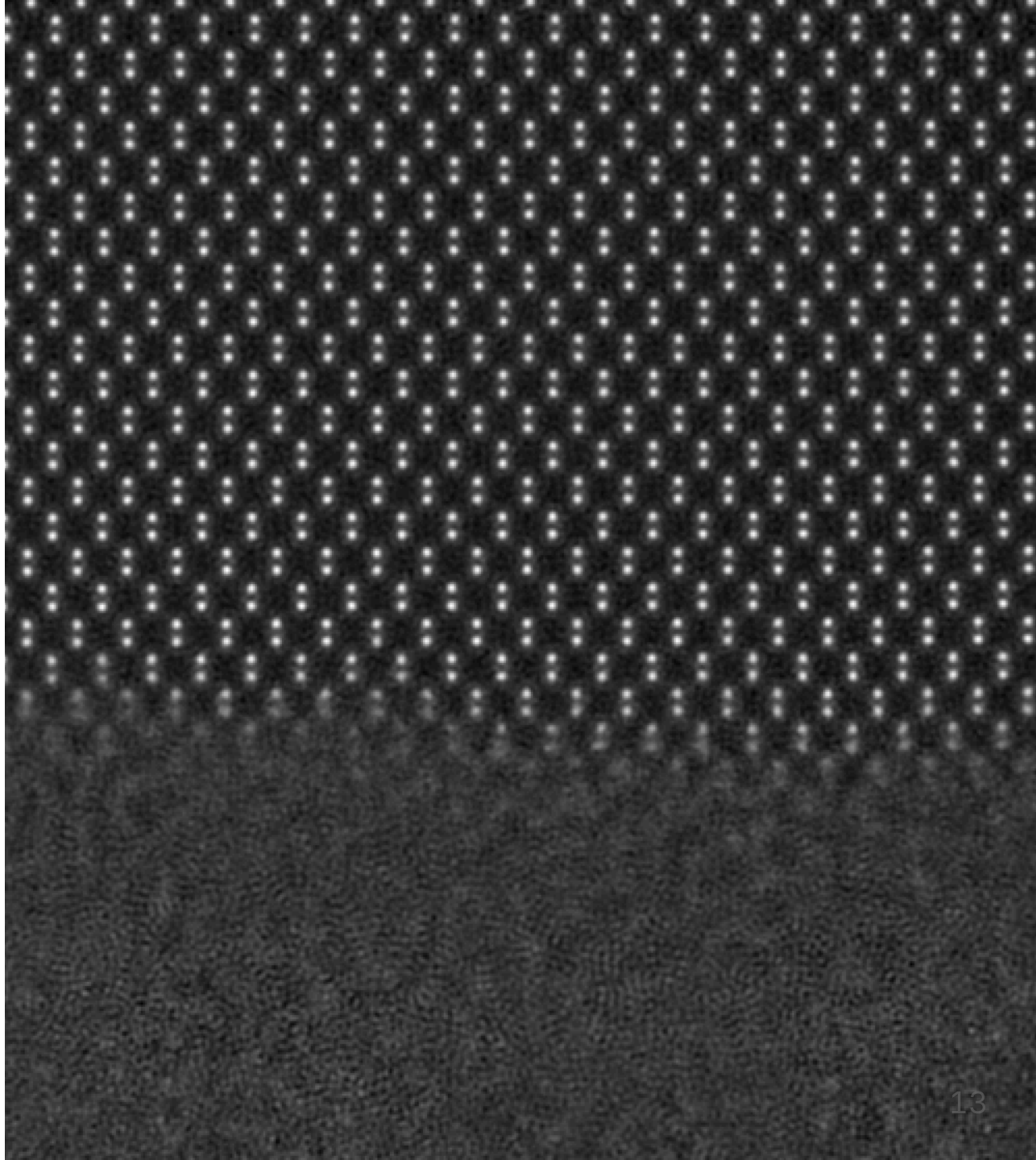

b
Unmasked Ronchigram
Mask
Bright Field <30 mrad
or
Annular Dark Field >30 mrad
or
Annular Bright Field >15-30 mrad
c
$DyScO_3$
Dy
Sc
O
HAADF Detector
80-200 mrad
Universal Detector Reconstructed Images
Bright Field
Annular Dark Field
Annular Bright Field
d
e
f
g
5 Å
5 Å
5 Å
5 Å

Ptychography



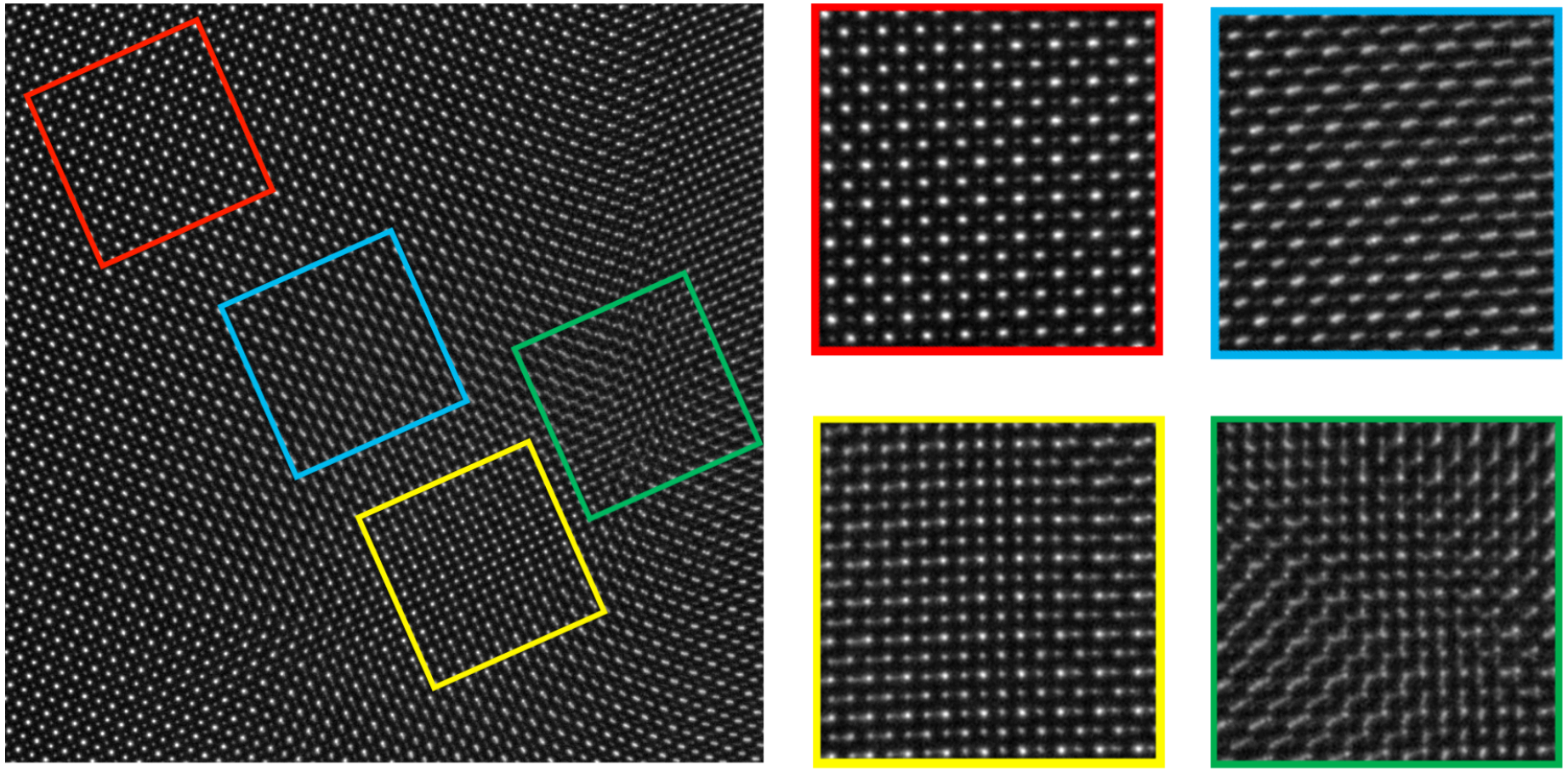
Examples

Depth-resolved 3D Ptychographic imaging of FIB-damaged Silicon



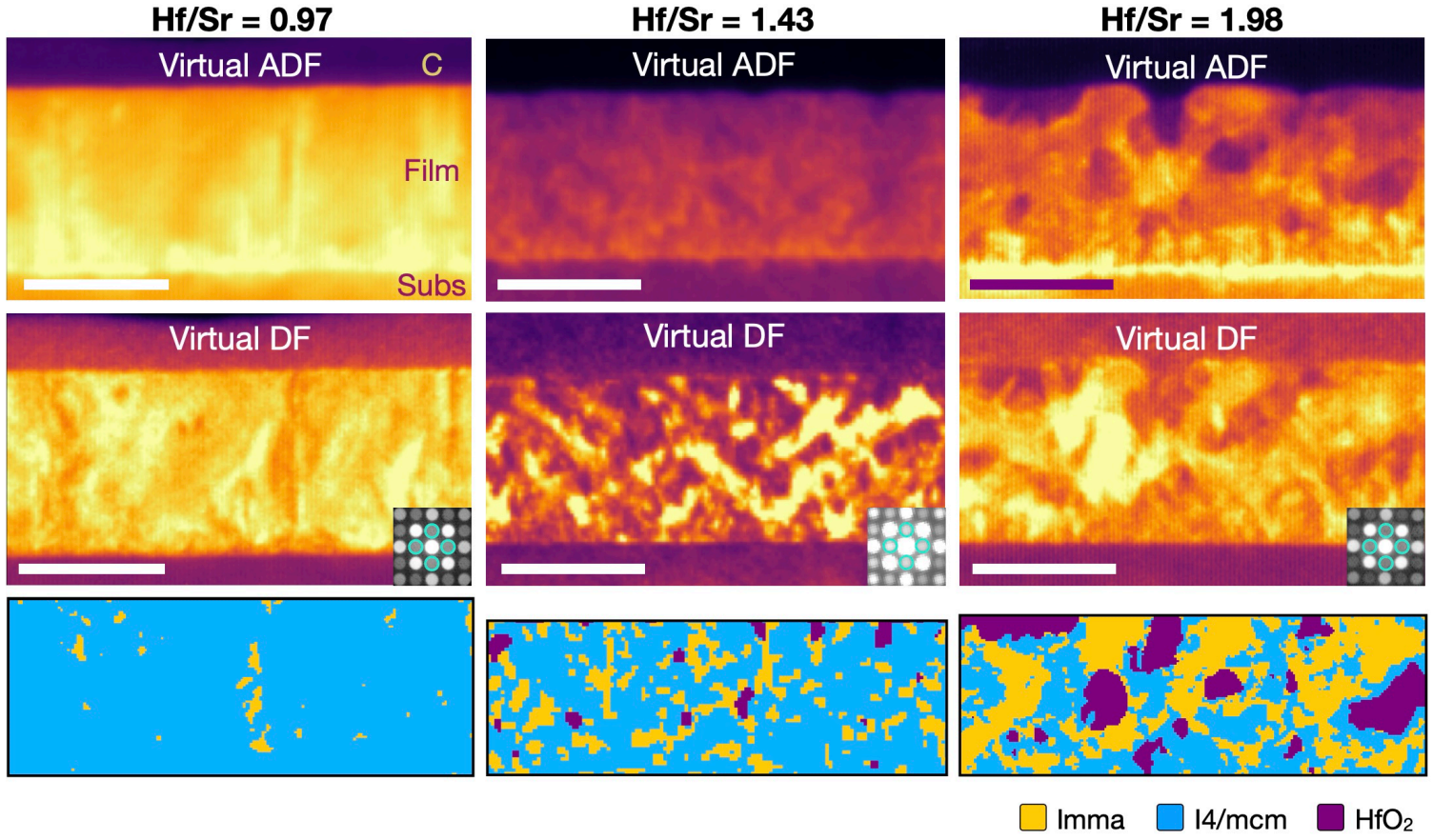
Examples

BaTiO3 dislocation analysis using 3D ptychography



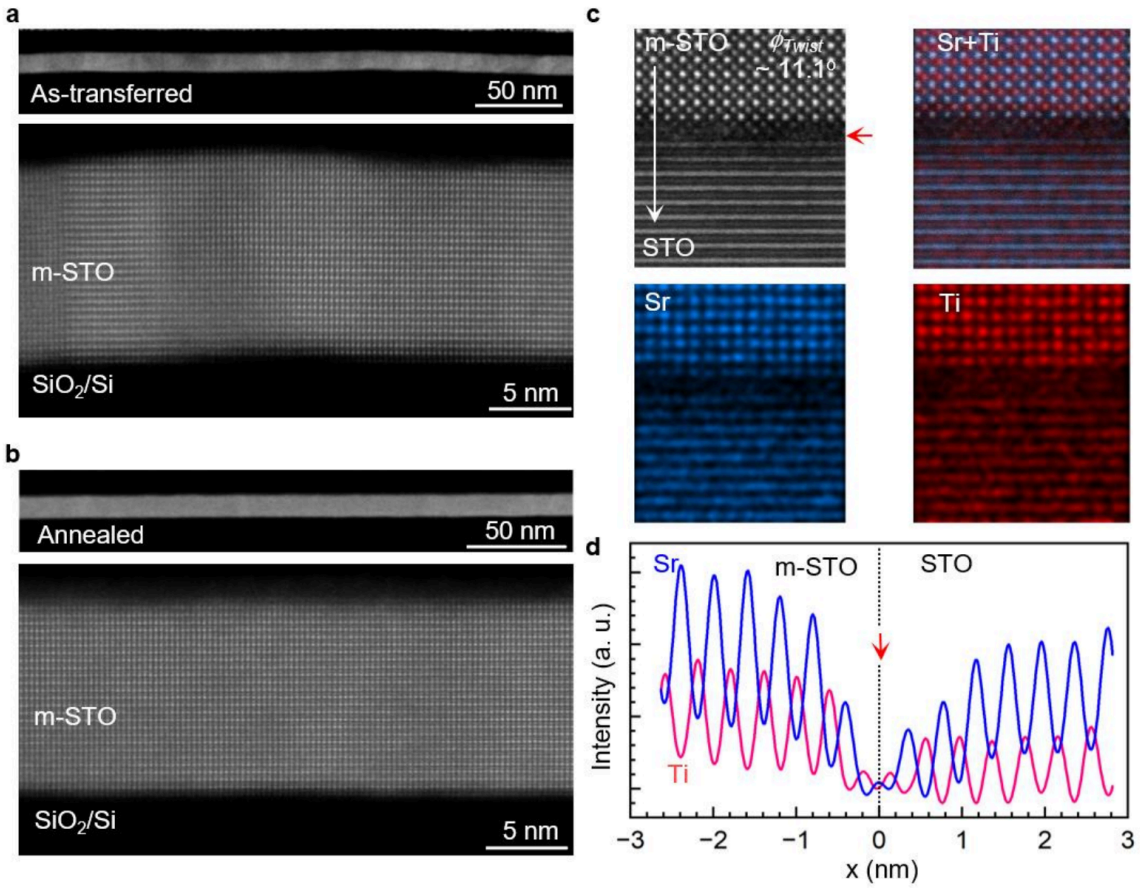
Examples

4D-STEM crystallographic phase mapping of SrHfO3



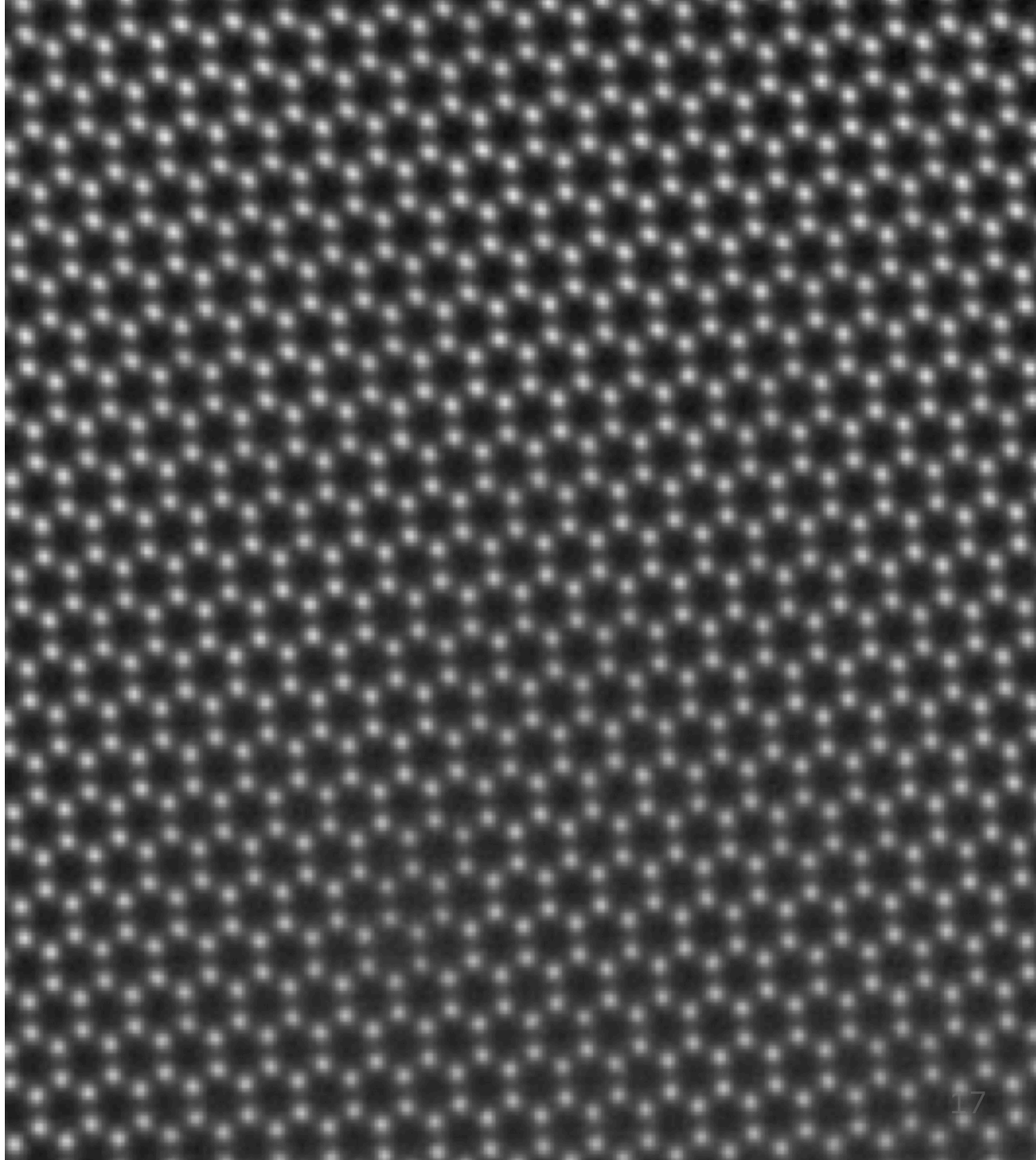
Examples

HAADF-STEM and EDS imaging of SrTiO3 interface



Examples

3D Ptychography of hBN using Themis-Z



Active topics

- Phase coexistance in high-κ materials
- Properties of high band gap oxides
- Chemical + structural analysis of oxides
- Ptychographic imaging of semiconductor heterostructure
- Ptychography algorithm improvements
- Device fabrication using perovskite oxide thin films, including FeFETs
- Strain engineering to develop novel ferroelectric perovskite oxides
- Freestanding membrane fabrication and transfer

Q & A